

Overview of Algorithmic bias

Subject: Algorithmic bias

1.

Algorithms are difficult to define, but may be generally understood as lists of instructions that determine how programs read, collect, process, and analyze data to generate a usable output. For a rigorous technical introduction, see Algorithms. Advances in computer hardware and software have led to an increased capability to process, store and transmit data. This has in turn made the design and adoption of technologies such as machine learning and artificial intelligence technically and commercially feasible. By analyzing and processing data, algorithms are the backbone of search engines, social media websites, recommendation engines, online retail, online advertising, and more. Contemporary social scientists are concerned with algorithmic processes embedded into hardware and software applications because of their political and social impact, and question the underlying assumptions of an algorithm's neutrality. The term algorithmic bias describes systematic and repeatable errors that create unfair outcomes, such as privileging one arbitrary group of users over others. For example, a credit score algorithm may deny a loan without being unfair, if it is consistently weighing relevant financial criteria. If the algorithm recommends loans to one group of users, but denies loans to another set of nearly identical users based on unrelated criteria, and if this behavior can be repeated across multiple occurrences, an algorithm can be described as biased. This bias may be intentional or unintentional (for example, it can come from biased data obtained from a worker that previously did the job the algorithm is going to do from now on).

2.

Technical bias emerges through limitations of a program, computational power, its design, or other constraint on the system. Such bias can also be a restraint of design, for example, a search engine that shows three results per screen can be understood to privilege the top three results slightly more than the next three, as in an airline price display. Another case is software that relies on randomness for fair distributions of results. If the random number generation mechanism is not truly random, it can introduce bias, for example, by skewing selections toward items at the end or beginning of a list. A decontextualized algorithm uses unrelated information to sort results, for example, a flight-pricing algorithm that sorts results by alphabetical order would be biased in favor of American Airlines over United Airlines. The opposite may also apply, in which results are evaluated in contexts different from which they are collected. Data may be collected without crucial external context: for example, when facial recognition software is used by surveillance cameras, but evaluated by remote staff in another country or region, or evaluated by non-human algorithms with no awareness of what takes place beyond the camera's field of vision. This could create an incomplete understanding of a crime scene, for example, potentially mistaking bystanders for those who commit the crime. Lastly, technical bias can be created by attempting to formalize decisions into concrete steps on the assumption that human behavior works in the same way. For example, software weighs data points to determine whether a defendant should accept a plea bargain, while ignoring the impact of emotion on a jury. Another unintended result of this form of bias was found in the plagiarism-detection software Turnitin, which compares student-written texts to information found online and returns a probability score that the student's work is copied. Because the software compares long strings of text, it is more likely to

identify non-native speakers of English than native speakers, as the latter group might be better able to change individual words, break up strings of plagiarized text, or obscure copied passages through synonyms. Because it is easier for native speakers to evade detection as a result of the technical constraints of the software, this creates a scenario where Turnitin identifies foreign-speakers of English for plagiarism while allowing more native-speakers to evade detection.

3.

Online hate speech In 2017 a Facebook algorithm designed to remove online hate speech was found to advantage white men over black children when assessing objectionable content, according to internal Facebook documents. The algorithm, which is a combination of computer programs and human content reviewers, was created to protect broad categories rather than specific subsets of categories. For example, posts denouncing "Muslims" would be blocked, while posts denouncing "Radical Muslims" would be allowed. An unanticipated outcome of the algorithm is to allow hate speech against black children, because they denounce the "children" subset of blacks, rather than "all blacks", whereas "all white men" would trigger a block, because whites and males are not considered subsets. Facebook was also found to allow ad purchasers to target "Jew haters" as a category of users, which the company said was an inadvertent outcome of algorithms used in assessing and categorizing data. The company's design also allowed ad buyers to block African-Americans from seeing housing ads. While algorithms are used to track and block hate speech, some were found to be 1.5 times more likely to flag information posted by Black users and 2.2 times likely to flag information as hate speech if written in African American English.

4.

Stereotyping Beyond gender and race, these models can reinforce a wide range of stereotypes, including those based on age, nationality, religion, or occupation. This can lead to outputs that homogenize, or unfairly generalize or caricature groups of people, sometimes in harmful or derogatory ways. A recent focus in research has been on the complex interplay between the grammatical properties of a language and real-world biases that can become embedded in AI systems, potentially perpetuating harmful stereotypes and assumptions. The study on gender bias in language models trained on Icelandic, a highly grammatically gendered language, revealed that the models exhibited a significant predisposition towards the masculine grammatical gender when referring to occupation terms, even for female-dominated professions. This suggests the models amplified societal gender biases present in the training data.

5.

Correlations Unpredictable correlations can emerge when large data sets are compared to each other. For example, data collected about web-browsing patterns may align with signals marking sensitive data (such as race or sexual orientation). By selecting according to certain behavior or browsing patterns, the end effect would be almost identical to discrimination through the use of direct race or sexual orientation data. In other cases, the algorithm draws conclusions from correlations, without being able to understand those correlations. For example, one triage program gave lower priority to asthmatics who had pneumonia than asthmatics who did not have pneumonia. The program algorithm did this because it simply compared survival rates: asthmatics with pneumonia are at the highest risk. Historically, for this same reason, hospitals typically give such asthmatics the best and most immediate care.